

Lab Session 6 (Open-Ended Lab)

Open Ended Lab	
Blooms Taxonomy	GAs
C3 P2	GA-2 GA-5

SUBMITTED TO: SIR RIZWAN AHMED

NAME: ALI SHEHZAN

REGISTRATION ID: 67158

Table of Contents

1. Title:.....	3
2. Motivation:	3
3. Concept:	3
4. Problem Statement:.....	4
5. Design / Ways and Means:.....	4
i. Input Module:	4
ii. Processing Module:	4
iii. Detection Engine:.....	4
iv. Machine Learning Module:	5
v. Scoring System:	5
vi. Output Module:.....	5
6. Analysis and Reporting:.....	5
7. Lab Activity:	6
8. Deliverables:.....	6
i. Background / Theory:.....	6
ii. Procedure / Methodology:	6
iii. Data Collection:	7
9. Flowchart / Block Diagram:	8
10. Analysis:.....	9
11. Results:.....	9
12. Discussion on Results:	9
13. Concluding Remarks:.....	9
14. References:	10

1. Title:

STAR Log Analyzer – AI and Machine Learning using Isolation Forest

2. Motivation:

With the rapid growth of digital systems, cybersecurity threats such as brute-force attacks, unauthorized logins, bot activity, and abnormal network behavior have become increasingly common. Organizations generate massive amounts of log data every second, making manual monitoring inefficient and unreliable.

This project is motivated by the need to automate log analysis and improve security monitoring using intelligent techniques. By combining rule-based detection with machine learning, the system aims to simulate real-world Security Information and Event Management (SIEM) systems used in modern Security Operations Centers (SOC).

3. Concept:

The system is based on a hybrid cybersecurity detection approach that combines:

- Rule-based threat detection (manual logic rules)
- Machine Learning-based anomaly detection (Isolation Forest algorithm)

The system processes log files, extracts meaningful features, and identifies suspicious patterns. It then classifies IP behavior into different risk levels such as Low, Medium, High, and Critical. This allows security analysts to quickly identify threats and take action.

4. Problem Statement:

The objective of this project is to design and implement a system that can automatically analyze network log files, detect malicious activities, and classify IP addresses based on their behavior using both rule-based logic and machine learning techniques. The system should also generate alerts for high-risk activity.

5. Design / Ways and Means:

The system is developed using Python and Streamlit and consists of the following modules:

Input Module:

Users upload a CSV log file containing:

- IP Address
- Timestamp
- Action (login, request, scan, etc.)
- Status (success or failed)

Processing Module:

The system reads the dataset using Pandas and organizes it based on IP addresses for analysis.

Detection Engine:

The system detects multiple types of cyber threats, including:

- Brute-force attacks (multiple failed login attempts)
- High traffic or unusual request volume
- Suspicious login behavior (failed attempts followed by success)
- Unknown IP activity (untrusted sources)
- Bot-like rapid activity patterns

Machine Learning Module:

An Isolation Forest algorithm is used to detect anomalies in IP behavior. It identifies unusual patterns that do not match normal traffic behavior.

Scoring System:

A risk score is calculated for each IP based on:

- Number of failed attempts
- Total activity level
- ML anomaly detection result

Output Module:

The system displays:

- Complete analysis table
- Security alerts for threats
- Safe IP list
- Risk classification results

6. Analysis and Reporting:

Each IP address is analyzed individually using both rule-based logic and machine learning outputs. The system evaluates:

- Frequency of failed login attempts
- Total number of actions performed
- Behavioral patterns over time
- ML-based anomaly detection result

Based on these factors, a risk score is calculated and each IP is classified into:

- Low Risk
- Medium Risk
- High Risk
- Critical Risk

This classification helps in identifying potential threats efficiently.

7. Lab Activity:

- Developed a Streamlit-based web application for log analysis
- Created synthetic log datasets for testing different attack scenarios
- Implemented rule-based cybersecurity detection logic
- Integrated machine learning model (Isolation Forest)
- Performed IP-wise behavioral analysis
- Generated real-time security alerts
- Displayed results in a structured dashboard format

8. Deliverables:

i. Background / Theory:

Cybersecurity log analysis is an essential part of SOC operations. Logs store detailed records of system and network activities. Machine learning enhances traditional monitoring by automatically detecting abnormal behavior without predefined rules.

ii. Procedure / Methodology:

- Upload CSV log file
- Load data using Pandas
- Extract features for each IP
- Apply rule-based detection logic
- Apply machine learning anomaly detection

- Calculate risk scores
- Classify IPs into risk levels
- Display results in dashboard

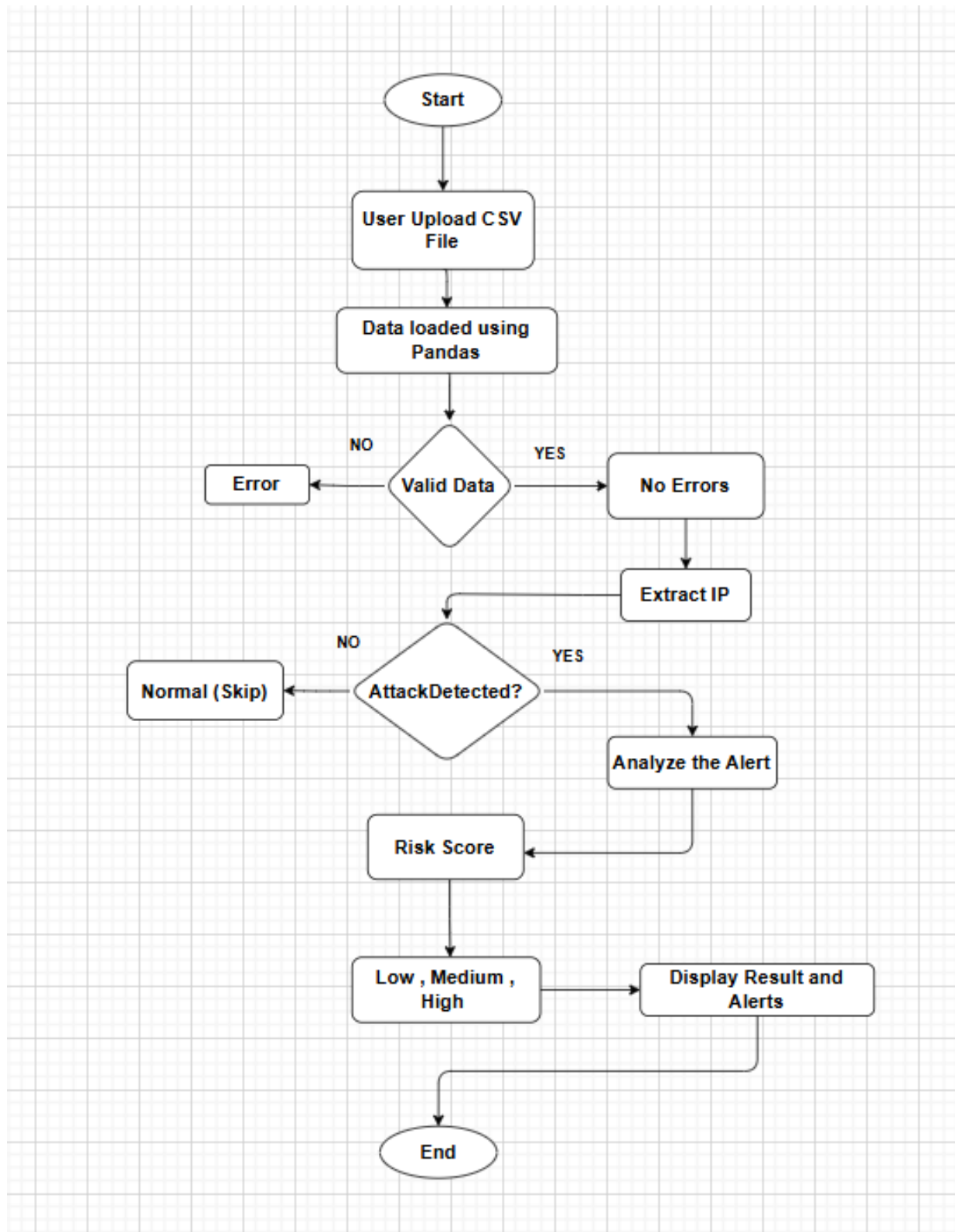
iii. Data Collection:

A synthetic dataset was created containing simulated network logs. Each record includes:

- IP Address
- Timestamp
- Action type
- Status (success/failure)

The dataset includes normal user behavior as well as attack scenarios such as brute-force attempts and bot-like activity.

9. Flowchart / Block Diagram:



10. Analysis:

The system analyzes each IP based on multiple parameters. Rule-based detection identifies known attack patterns, while machine learning detects hidden or unknown anomalies. This combination improves detection accuracy and reduces false positives.

11. Results:

The system successfully identifies:

- Brute-force attacks
- Suspicious login behavior
- Bot-like rapid activity
- High traffic anomalies
- Normal safe user behavior

It effectively classifies IPs into different risk levels and highlights critical threats.

12. Discussion on Results:

The hybrid approach of combining machine learning with rule-based detection provides a more accurate and realistic cybersecurity solution. Rule-based methods detect known threats, while machine learning identifies unusual patterns that may not follow predefined rules. This makes the system suitable for real-world SOC environments.

13. Concluding Remarks:

This project demonstrates an intelligent cybersecurity log analysis system capable of detecting and classifying network threats. It simulates real SOC operations by analyzing logs, identifying anomalies, and generating alerts. The system can be further enhanced by integrating real-time streaming data and advanced deep learning models.

14. References:

- Stallings, W. – Network Security Essentials
- NIST Cybersecurity Framework: <https://www.nist.gov/cyberframework>
- Python Pandas Documentation: <https://pandas.pydata.org/docs/>
- Scikit-learn Documentation: <https://scikit-learn.org/>
- Streamlit Documentation: <https://docs.streamlit.io>